

Correlated electronic states of C_{60} fragments[☆]

Aparna Chakrabarti^a, Y. Anusooya^a, S. Ramasesha^{a,b,*}

^a*Solid State and Structural Chemistry Unit, Indian Institute of Science, Bangalore 560 012, India*

^b*Jawaharlal Nehru Centre for Advanced Scientific Research, Indian Institute of Science Campus, Bangalore 560 012, India*

Received 24 February 1995; accepted 15 March 1995

Abstract

We have studied electronic states of various fragments of C_{60} within the Pariser–Parr–Pople (PPP) model and have obtained structural, magnetic and spectral properties of these molecules. The fragments studied include corannulene, fluoranthene and pyracylene. Pyracylene is studied using the exact valence bond (VB) approach while fluoranthene and corannulene are studied using a novel restricted CI technique which employs molecular orbitals for constructing the VB functions. The electronic excitations, bond order and ring currents are calculated for these systems. From these studies, the wide range of absorptions in C_{60} can be viewed as those localized on pyracylene units or on the corannulene/fluoranthene units. The bond orders and ring currents show the hexagons to be similar to benzene rings. The pentagon–hexagon bonds are also found to be longer than the hexagon–hexagon bonds.

1. Introduction

Buckminsterfullerene or C_{60} , the recently discovered [1,2] soccer-ball shaped third form of carbon has attracted a great deal of experimental and theoretical attention. Although interesting structural phase transitions have been observed in solid C_{60} , the truly remarkable properties associated with this system are electronic in nature. Consequently, theoretical electronic structure studies on this system have come to the fore [3,4].

Each of the carbon atoms in C_{60} are three coordinated, although the hybridization on the carbon atoms is not strictly sp^2 and it consists of pentagons and hexagons. Relatively short bonds (1.391 Å) are shared between two hexagons and the longer bonds

(1.455 Å) are shared between a pentagon and a hexagon [5]. C_{60} can be described as a truncated icosahedron with the truncating planes being parallel to each of the interior pentagonal planes of the icosahedron.

Doped and undoped C_{60} show a variety of interesting properties. When inter-calated with alkali elements, C_{60} becomes superconducting with transition temperatures (T_c) in the range of 18–40 K depending on the dopant [6]. This observation has led to considerable interest in the electronic structure and the mechanism of superconductivity in this system. There has also been evidence for soft ferromagnetism [7,8] in the 1 : 1 adduct, C_{60} –TDAE. The TDAE (tetrakis dimethyl amino ethylene) molecule donates an electron to C_{60} resulting in the charge-transfer solid C_{60} –TDAE. The d.c. susceptibility measurements showed that at 16 K the material becomes a soft itinerant ferromagnet [7].

The conventional electron–phonon mechanism

[☆] Contribution no. 1092 from Solid State and Structural Chemistry Unit.

* Corresponding author.

for superconductivity in this system has been investigated by several workers [9,10]. However, this mechanism has failed to explain several experimental observations. For example, the value of $2\Delta/k_B T_c = 3.5$ from BCS (Bardeen–Cooper–Schrieffer) theory does not agree with the experimental value of 5.03 ± 0.02 (K_3C_{60}) and 5.20 ± 0.30 (Rb_3C_{60}) [11]. There are also difficulties in explaining Raman data on doped C_{60} since the $q \approx 0$ optical phonons cannot easily be broadened by electron–phonon scattering (at least at the level of simple Fermi–Golden rule estimates) as the kinematic constraints $\omega = (\epsilon_{k+q} - \epsilon_q)$ cannot be satisfied [12].

The above experimental observations have forced one to explore the possibility of a purely electronic mechanism for superconductivity in these systems [13]. In recent years, there have been rigorous electronic structure calculations of the ground state properties of fullerenes. These include self-consistent field calculations using STO-6-31G* basis [14], MNDO [15] calculations on carbon clusters and calculations employing linear muffin-tin orbitals [16]. Most of these calculations address themselves to questions related to ground state geometry, resonance stabilization, prediction of magic numbers, prediction of vibrational spectra and “in-cage” versus “out-cage” doping with alkali ions. There is, however, a perturbative calculation [4], within the one-band Hubbard model in doped C_{60} which indicates the possibility of Cooper pair formation in this system.

While the above studies are important in themselves, they cannot, by the very nature of the approximations, provide reliable excitation gaps, spin and charge density distributions and correlation functions which are important for understanding properties such as ferromagnetism and superconductivity in the fullerenes [7,17,18]. Moreover, fullerenes are very similar to other conjugated organic systems. Although the hybridization on the carbon atom in the fullerenes is not strictly sp^2 , the three coordinated carbon atoms, in fullerenes, allow theoretical treatment similar to those in the conventional homoorganic systems. In fact, several theoretical studies have predicted C_{60} to have a degree of aromaticity close to that of benzene ([19] and the references therein).

In conjugated systems, the discrepancy between the self-consistent field results and the experiments have been satisfactorily resolved by explicit inclusion of electron correlation. Excellent agreement between theory and experiment has been obtained in the case of polyenes and other conjugated systems. Hence, any reliable correlated electronic structure calculations for C_{60} should include explicit electron correlations. Minimal quantum cell models, such as Hubbard, PPP Pariser–Parr–Pople models [20,21], are routinely used for electronic structure calculations of organic conjugated systems in the literature [22]. These models span very large Hilbert spaces even for relatively small systems. A typical dimensionality of the space is a few million for a 14 site Hubbard model. Hence, exact solution for a single C_{60} molecule within any of these models, though desirable, is impossible in the foreseeable future.

Approximate yet reliable solution from a novel CI technique, namely the MO-VB scheme [23] has been obtained for many conjugated systems. Within the PPP model it gives good vertical energy excitation spectra and microscopic properties. Good agreement between approximate MO-VB studies and exact solutions within Hubbard and PPP models [23,24] for large molecules with up to 14 π electrons has prompted us to investigate the electronic structure of C_{60} and its basic fragments. While even within this scheme the full C_{60} molecule is currently inaccessible, because of the large dimensionality (11 546 644 for singlets for threshold equal to band width) and relative lack of sparseness (unlike

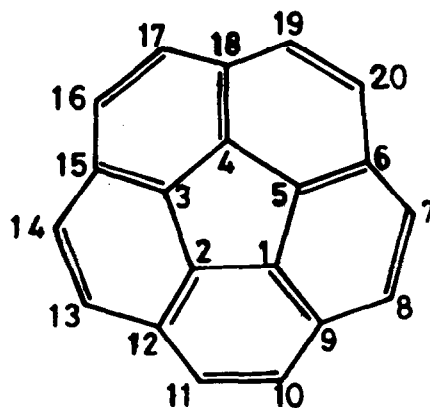


Fig. 1. Structure of corannulene. The geometry is assumed to have the same curvature as that of the bucky ball.

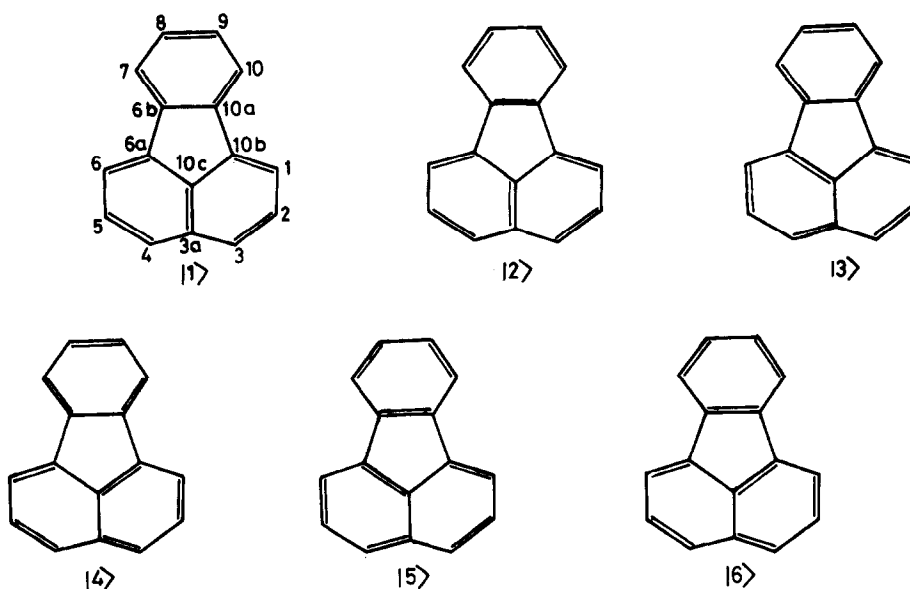


Fig. 2. Possible covalent Kekulé structures for fluoranthene.

in AO-VB methods), we can still reliably study basic fragments of C_{60} , employing this scheme.

Our studies on various fragments of C_{60} are aimed at obtaining geometrical and spectral properties. The fragments studied include a 20 carbon system (corannulene, Fig. 1), a 16 carbon system (fluoranthene, Fig. 2) and a 14 carbon system (pyracylene, Fig. 3). While exact AO-VB calculations on corannulene and fluoranthene are computationally out of reach, the pyracylene system can be studied exactly by the AO-VB method. The fluoranthene and corannulene systems have been studied using the approximate MO-VB method. In what follows, we present a brief outline of the model and techniques in section 2 and in section 3, our results on these systems are presented and discussed in the context of C_{60} .

2. The computational technique

2.1. PPP Model Hamiltonian

The PPP model is obtained from a many-body Hamiltonian involving only π electrons by making a zero differential overlap approximation. The actual Hamiltonian is given by

$$H_{\text{PPP}} = \sum_{\langle i,j \rangle} \sum_{\sigma} t_{ij} (a_{i\sigma}^{\dagger} a_{j\sigma} + h.c.) + U/2 \sum_i \hat{n}_i (\hat{n}_i - 1) + \sum_{i < j} V_{ij} (\hat{n}_i - z_i) (\hat{n}_j - z_j) \quad (1)$$

where the first term corresponds to the one-electron

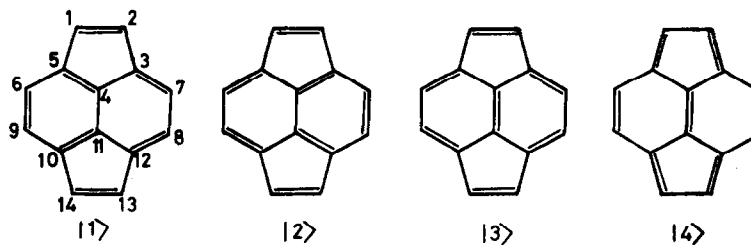


Fig. 3. Possible covalent Kekulé structures for pyracylene.

part of the Hamiltonian and involves electron transfer between π orbitals on bonded atoms. The second term corresponds to electron repulsion between electrons in the same orbital and the third term to repulsions between electrons in different orbitals. In the above Hamiltonian, t_{ij} is the transfer integral between bonded pairs of atoms and the diagonal part of t_{ij} corresponds to the site energy of the orbitals which in an alternant hydrocarbon is presumed equal and can be used to define the zero of energy. The operator $a_{i\sigma}^\dagger$ ($a_{i\sigma}$) creates (annihilates) an electron with spin σ in the i th orbital, $\hat{n}_i$ is the corresponding occupation number operator and z_i is the occupancy of the orbital i that leaves the site neutral. The parameters U and V_{ij} correspond to on-site and inter-site electron repulsions. If the inter-site electron repulsions are neglected then we regain the familiar Hubbard model. In the PPP model, the on-site electron repulsion U is set to 11.26 eV which is the difference of ionization energy and electron affinity for carbon, and V_{ij} is interpolated between U for $r_{ij} = 0$ and e^2/r_{ij} for $r_{ij} \rightarrow \infty$ using the Ohno interpolation [25],

$$V_{ij} = 14.397 \{ [28.794 / (U_i + U_j)]^2 + r_{ij}^2 \}^{-1/2} \quad (2)$$

The PPP model Hamiltonian conserves total spin. So using spin-adapted basis functions would greatly reduce the computational effort, besides giving the correct spin label to the states. We have employed a diagrammatic VB basis in our computation. The diagrammatic VB basis is easy to construct and manipulate since the VB diagrams can be represented uniquely as an integer in the bit representation scheme [22]. The VB basis although complete and linearly independent is non-orthogonal and the resulting Hamiltonian matrix is non-symmetric. However, there exist efficient small matrix algorithms that allow solutions to low-lying states of large, sparse, non-symmetric matrices [26].

2.2. Valence bond method in molecular orbital basis

Exact VB theory for large systems is impossible since the number of VB states in the complete basis increases rapidly with system size. Approximations

for restricting the size of the basis, therefore, becomes essential for dealing with large systems. The two-electron part of the PPP Hamiltonian is diagonal in the AO-VB basis and it appears to be quite straightforward to eliminate VB states which have two-electron repulsion energies above a chosen threshold. However, comparison of results from such a restricted CI scheme with exact results for small systems shows that a significant fraction of the total number of states have to be retained for reasonable accuracy.

Another way of restricting the basis is to diagonalize the one-electron part of the Hamiltonian and eliminate those configurations whose total one-electron energy exceeds a chosen threshold. This is the familiar restricted MO-CI technique. There are two ways in which this is most commonly practised. In one, known as the complete active space CI, a truncated set of occupied and unoccupied MOs replace the full set and all possible configurations in this truncated one-electron basis are used in a CI scheme. In another version, the restriction is imposed by curtailing the possible number of particles and holes involved in generating excited configurations. Typically this cut-off is placed at 1, 2, 3 or 4 particle-hole excitations leading to “singles”, “doubles”, “triples” or “quadruples” CI. In the context of Hubbard and the PPP models it is found that such a CI scheme does not give the correct nature of excited states over a wide range of the Hubbard parameter U . For small U , the lowest excitation should be charge-like and for large U , it should be spin-like.

We have recently experimented with a third type of CI scheme in the MO space [23]. In this scheme, we allow for all possible particle-hole excitations with the only restriction that the total one-electron energy with reference to the ground state one-electron energy be below a chosen threshold. When the threshold is chosen to be the total spread in the MO spectrum (referred to as band width) we could have one, two or many particle-hole excitations although none of these types of excitations (except the one particle-hole) need be completely exhausted. Our studies on a wide range of molecules have shown [23,24] that such a scheme reproduces the properties of the ground and low-lying states to reasonable accuracy.

In the computational technique employed by us, VB functions over all configurations below a given threshold are obtained for the given total spin. The procedure involves Lowdin orthogonalization of the MO functions and employs Rumer–Pauling rules to construct the VB functions in the MO basis. Hence we call this the MO-VB technique. The PPP Hamiltonian in the MO basis can be written as

$$H = \sum_{i\sigma} \epsilon_i b_{i\sigma}^\dagger + 1/2 \sum_{ijkl} V_{ijkl} (\hat{E}_{ij} \hat{E}_{kl} - \delta_{jk} \hat{E}_{il}) - \sum_{ij} V_{ij} \hat{E}_{ij} + V \quad (3)$$

$$\hat{E}_{ij} = \sum_{\sigma} b_{i\sigma}^\dagger b_{j\sigma} \quad (4)$$

where $b_{i\sigma}^\dagger$ ($b_{i\sigma}$) corresponds to creation (annihilation) of an electron in the i th molecular orbital with spin σ and ϵ_i is the energy of the i th MO. V_{ijkl} , V_{ij} and V are defined as

$$V_{ijkl} = \sum_{\mu\nu} c_{\mu i} c_{\mu j} c_{\nu k} c_{\nu l} V_{\mu\nu} \quad (5)$$

$$V_{ij} = \sum_{\mu\nu} (c_{\mu i} c_{\mu j} + c_{\nu i} c_{\nu j}) V_{\mu\nu} \quad (6)$$

$$V = \sum_{\mu\nu} V_{\mu\nu} \quad (7)$$

where $c_{\mu i}$ terms are the coefficients of the μ th AO in the i th MO and all the summation runs over the entire AO basis. $V_{\mu\nu}$ terms are the two-center integrals in the AO-VB basis which are parametrized as discussed earlier, and for $\mu = \nu$ reduce to Hubbard U_μ . The Hamiltonian matrix is non-symmetric and much less sparse than in the AO-VB technique. However, the dimensionality of the MO-VB basis is far smaller and the loss of sparseness is not a serious problem for systems with ≤ 24 orbitals. The generation of the Hamiltonian matrix is carried out as in the AO-VB case. However, the interaction part is off-diagonal and involves very general two-electron integrals in the MO basis. These can be handled by resolving the “illegal” VB diagrams into “legal” (those obeying Rumer–Pauling rules) diagrams by applying the simple rules discussed in Ref. [22].

Computation of properties involves inverse

transformation of the eigenstates in the MO-VB basis to the AO-VB basis [24]. Our scheme of calculations, besides being accurate for the low-lying eigenstates, also has the advantage of providing automatic spin labeling of the states. However, spatial symmetries have so far not been incorporated in this scheme.

2.3. Calculation of properties

Computation of most properties such as transition dipoles and bond order are well documented [22]. However, computation of ring currents which we have carried out on these systems is less well documented. Exact ring currents of correlated models were first obtained by Kuwajima and Soos [27]. The ring currents are directly related to the second order shift in energy ($E^{(2)}$) in an applied magnetic field and it is calculated by implicitly summing over all the excited states [27] that the chosen VB manifold spans. The second order energy shift is given by,

$$E^{(2)}(\text{PPP}) = \langle G | v_+ | G \rangle + 2 \sum_{R \neq G} \frac{|\langle G | v_- | R \rangle|^2}{E_G - E_R} \quad (8)$$

where E_G and $|G\rangle$ are the ground state energy and eigenfunctions, E_R and $|R\rangle$ similarly are the energies and eigenstates of the excited states in the absence of perturbation. The operators v_+ and v_- are given by,

$$v_{\pm} = \sum_{k\sigma} \beta (a_{k\sigma}^\dagger a_{k+1\sigma} \pm a_{k+1\sigma}^\dagger a_{k\sigma}) \quad (9)$$

and the second term is evaluated exactly by obtaining

$$|\psi'\rangle = \sum_{R \neq G} \frac{\langle R | v_- | G \rangle}{E_R - E_G} |R\rangle \quad (10)$$

by solving the equation,

$$(H - E_G) |\psi'\rangle = v_- |G\rangle \quad (11)$$

as a matrix equation. The second order correction to energy is given by,

$$E^{(2)} = \langle G | v_+ | G \rangle + 2 \langle \psi' | v_- | G \rangle \quad (12)$$

$E^{(2)}$ is proportional to ring current if the area S is proportional to N^2 (N is the number of sites) and

Table 1

Excitation energies (in eV) for pyracylene (AO-VB), fluoranthene (MO-VB), corannulene (MO-VB) and C₆₀. Experimental values are given in parentheses. Experimental intensities are given as log ϵ . For C₆₀, only experimental energies are reported [38]

Pyracylene			Fluoranthene			Corannulene	C ₆₀
Label	Energy	Osc. Strength	Label	Energy	Osc. Strength	Energy	Energy
1 ¹ B	1.95	0.0	1 ¹ A ₁	3.88 (3.41)	– (0.17)	4.18	(1.98) (2.07)
2 ¹ A	2.20	0.0	2 ¹ B ₂	4.35 (3.84)	– (0.05)	–	(2.29) (2.48)
3 ¹ A	2.75	0.0	2 ¹ A ₁	4.50 (4.30)	– (0.55)	–	(3.07) (3.77)
2 ¹ B	3.49 (3.05 ± 0.15)	0.05	–	–	–	–	(4.82) (5.82)
3 ¹ B	4.34 (3.70 ± 0.10)	0.0	–	–	–	–	

for a field H , ring current I is given by,

$$I/H = (4\pi^2 e^2 / h^2 c) (S/N^2) E^{(2)} \quad (13)$$

In our studies we have used $E^{(2)}$ as a measure of the ring current since S and N^2 are fixed for a given molecule in a given geometry.

3. Results and discussion

In Table 1 are given the energy levels of the low-lying singlet and triplet excited states relative to their respective ground states for all the three fragments and C₆₀ (experimental result). In the case of pyracylene, the results are obtained from diagonalization of the full CI matrix of the PPP model while for other systems, truncated MO-VB basis has been employed.

Table 2

Bond order for the ground state and relative π bond orders ($\Delta p_{ij} = p_{ij}^x - p_{ij}^0$) for the unique bonds in the excited singlet states of pyracylene. Numbering of sites corresponds to Fig. 3

State	1–2	2–3	3–4	5–6	6–9	4–11
1 ¹ A	0.816	0.329	0.455	0.697	0.526	0.604
1 ¹ B	–0.198	0.210	–0.030	–0.183	0.166	–0.001
2 ¹ A	–0.482	0.369	–0.109	–0.275	0.227	0.108
3 ¹ A	–0.179	0.136	0.016	–0.172	0.102	–0.302
2 ¹ B	–0.133	0.104	–0.030	–0.190	0.076	–0.002
3 ¹ B	–0.309	0.074	–0.032	–0.216	0.128	–0.007

3.1. Spectral results and equilibrium geometries of fragments of C₆₀

In pyracylene, there are at least three singlet excitations below the lowest optically allowed excitation (oscillator strength 0.05) within the PPP model. In a real situation, symmetry breaking due to non-identical environments around the atoms, can lead to weak one-photon absorption to these states. The theoretical one-photon allowed absorptions are at slightly higher energies than the reported experimental values [28], as is to be expected for calculations on isolated molecules. The lowest triplet is 1.17 eV above the ground state. However, the phosphorescence data is not available (to our knowledge) for comparison. The lowest triplet–triplet absorption is at 0.97 eV which would be in the IR region. However, a 1.69 eV triplet–triplet absorption is also predicted from our calculations. The ground state geometry of pyracylene from bond order data (Table 2) shows that pyracylene has very strong ethylenic units on the pentagons. These are weakly bridged to a naphthalenic unit. The naphthalenic unit is also distorted with rather long pentagon–hexagon bonds.

The bond order data in the excited states (Table 2) show clearly that the geometries are significantly distorted from the equilibrium geometry of the naphthalene core connected by two vinyl bridges. The double and single bond characters

Table 3

Energy (eV) and relative π bond orders ($\Delta p_{ij} = p_{ij}^X - p_{ij}^G$) for the unique bonds and spin densities (ρ_s) for unique sites in the triplet states of pyracylene. Numbering of sites corresponds to Fig. 3

State	Energy	1–2	2–3	3–4	5–6	6–9	4–11	ρ_s^1	ρ_s^3	ρ_s^4	ρ_s^6
1 3A	1.17	–0.248	0.187	–0.051	–0.135	0.117	0.055	0.25	0.12	0.06	0.10
1 3B	2.13	–0.322	0.092	–0.005	–0.065	0.004	–0.021	0.34	0.03	0.04	0.12
1 3A	2.32	–0.039	0.024	0.014	–0.213	0.149	–0.046	0.09	0.31	–0.02	0.12
2 3B	2.86	–0.072	0.106	0.008	–0.127	–0.032	–0.251	0.07	0.10	0.23	0.22

are affected resulting in a rather weak vinyl bridge. The naphthalene core also changes and the fourth resonant Kekulé structure (Fig. 3) has a relatively dominant contribution in the excited state geometry compared with the ground state. The geometries in the excited states of A space shown more changes in the bond order pattern. The 4–11 bond retains its character except in the 2nd excited state in the A space (2^1A).

The geometry of triplet excited states resemble the ground state geometry with a moderate distortion as can be seen from the bond order data (Table 3). The strong double bond character of the 1–2 bond is weakened in the 1 3A and 1 3B states. The 4–11 bond is unaffected except in the 2 3B state. The spin density values in the triplet (Table 3) are consistent with the bond order data. Site 1 has a higher spin density in the first two excited states while site 4 shows a higher spin density value in the 2 3B state.

MO-VB studies on fluoranthene show singlet excited states at 3.88, 4.35 and 4.50 eV. The bond order pattern in this molecule in the ground state corresponds to delocalized benzene and naphthalene units connected by pentagons with rather long bonds. It is interesting to note that a 3.7 eV absorption is predicted for an isolated naphthalene molecule while a 4.23 eV absorption is predicted for benzene. Thus the fluoranthene molecule could be viewed as weakly coupled benzene and naphthalene molecules. The experimental spectra [29] are red-shifted by about 0.5 eV, if we assign the state at 3.88 eV to 1A_1 and the state at 4.35 eV to 1B_2 .

The bond order data in the excited states (Table 4) predict the 6–6a bond to have weaker double bond character. The 6a–6b bond has an increased double bond character in all the excited states

except the second excited singlet state. We observe that the most affected bonds are the 7–6b, 6a–6b and 6–6a bonds in all the excited states. The geometries are significantly distorted in the excited states from the equilibrium geometry of the naphthalene core connected to a benzene unit by a vinyl bridge.

MO-VB studies of corannulene are rather limited and the lowest excitation is predicted to be at 3.93 eV for uniform bond lengths. Introduction of longer hexagon–pentagon bonds leads to a blue shift in the absorption to 4.18 eV. The hexagon–hexagon bonds have a bond order (0.62) which is nearly the same as in benzene (0.69). Hexagon–pentagon bond order is lower (0.46) while the outer hexagon bonds have strong ethylenic character (bond order ≈ 0.79). The hexagon–hexagon and hexagon–pentagon bond order data is consistent with experimental bond lengths for the C_{60} molecule. The bond length

Table 4

Bond order for the ground state and relative π bond orders ($\Delta p_{ij} = p_{ij}^X - p_{ij}^G$) for the unique bonds in the excited singlet states of fluoranthene. Numbering of sites corresponds to Fig. 2

Unique bond	States				
	S ₀	S ₁	S ₂	S ₃	S ₄
10–10a	0.679	–0.111	–0.004	–0.119	–0.111
9–10	0.596	–0.033	–0.010	–0.111	–0.001
8–9	0.708	–0.050	0.014	–0.003	–0.055
6a–6b	0.326	0.135	0.045	0.114	0.176
6–6a	0.716	–0.170	–0.136	–0.119	–0.199
5–6	0.547	0.086	–0.069	0.069	0.117
4–5	0.757	–0.092	–0.121	–0.068	–0.126
3a–4	0.497	0.012	0.035	0.007	0.019
3a–10c	0.599	–0.025	–0.236	–0.004	–0.053
6b–10a	0.532	–0.097	–0.019	–0.083	–0.127
6a–10c	0.479	0.006	0.027	–0.044	–0.052

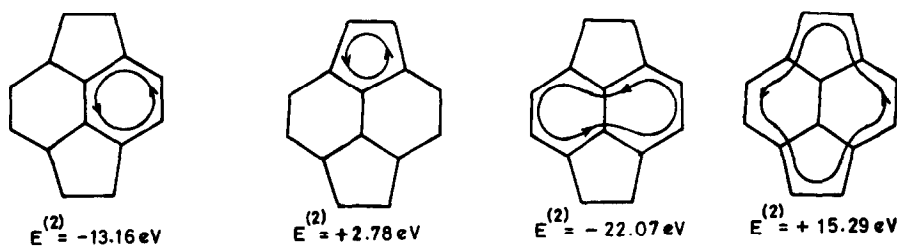


Fig. 4. $E^{(2)}$ values for different circulations in pyracylene. Directions of circulations are arbitrary.

obtained for the shorter bond in C_{60} is 1.39 Å from X-ray diffraction [5] and 1.40 Å from electron diffraction [30] while the longer bond length is 1.46 Å from both techniques. Coulson's formula [31,32] using our bond orders yields a value of 1.41 Å for the shorter bond and 1.45 Å for the longer bond in corannulene. The hexagon–pentagon and hexagon–hexagon bonds in pyracylene and fluoranthene are also similar to those in corannulene. It is interesting to note that the optical absorption of C_{60} spans a wide energy range (Table 1). It appears that the low-energy excitations are localized on pyracylene type fragments while the high energy excitations involve fluoranthene and corannulene type fragments.

3.2. Aromatic ring current in fragments of C_{60}

All the aromatic ring current results are analyzed in terms of $E^{(2)}$, the second order correction to energy due to the external magnetic field, relative to the $E^{(2)}$ values for benzene (−17.78 eV) and naphthalene (−26.14 eV). The ring current data on pyracylene (Fig. 4) show interesting variations. The benzene rings show large aromatic ring current ($E^{(2)} = -13.6 \text{ eV}$) while the pentagons show very small paramagnetic circulations ($E^{(2)} = +2.78 \text{ eV}$).

The naphthalenic circulation is diamagnetic and rather large (−22.07 eV) while the circulation along the perimeter of the molecule is paramagnetic ($E^{(2)} = +15.29 \text{ eV}$). Earlier theoretical calculations also predict a paramagnetic behavior of the pyracylene molecule [33,34]. The ring current results establish that pyracylene can be viewed as a $4n$ -annulene system with internal vinyl cross links. There are two types of protons in pyracylene. Those on the five-membered ring are expected to show an upfield shift in NMR due to paramagnetic ring current while the protons associated with six-membered rings are expected to be downfield shifted with respect to the other protons due to cancelations of the naphthalenic and benzenic diamagnetic ring currents with the paramagnetic ring current associated with the perimeter circulation. Comparison of this with our ring current calculation on acenaphthylene shows that all the rings as well as the perimeter have diamagnetic circulations in the six-membered rings. However, the circulation in the five-membered ring is very weakly diamagnetic ($E^{(2)} = -0.36 \text{ eV}$) in acenaphthylene.

The ring current in the fluoranthene molecule (Fig. 5) indicates that the six-membered ring is very much like benzene ($E^{(2)} = -16.61 \text{ eV}$) and

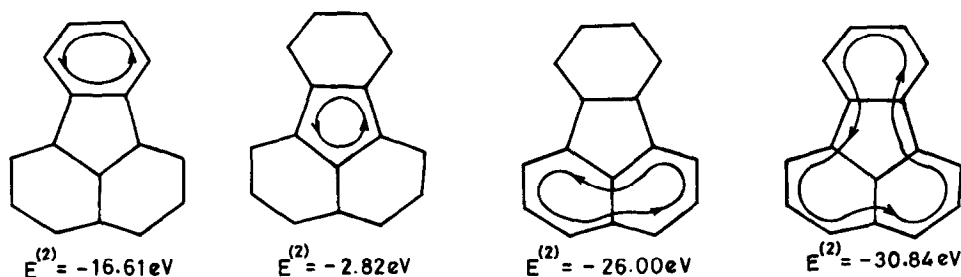


Fig. 5. $E^{(2)}$ values for different circulations in fluoranthene. Directions of circulations are arbitrary.

the naphthalenic unit has an $E^{(2)}$ value of -26.00 eV. These values are very close to the ring currents in isolated benzene and naphthalene molecules. The circulation in the five-membered ring bridging the two units is weakly diamagnetic. This may imply a slightly negative charge cloud on the five-membered ring. The perimeter circulation in the molecule is also diamagnetic with an $E^{(2)}$ value of -30.84 eV.

The ring current along the perimeter of the five-membered ring in corannulene is very small ($E^{(2)}$ is -6.11 eV) and that for a benzenic unit ($E^{(2)}$ is -12.76 eV). The conjugation seems to be large in the perimeter of the six-membered rings as indicated by the large $E^{(2)}$ value along the naphthalenic, phenanthrenic and 3,4-benzphenanthrenic units ($E^{(2)} = -22.29, -32.17$, and -42.31 eV, respectively). $E^{(2)}$ values in all the above mentioned units and along the perimeter ($E^{(2)} = -32.50$) indicate a diamagnetic nature of the corannulene molecule.

Recently π -electron ring current magnetic susceptibilities and endohedral chemical shifts of the fullerenes have been calculated with the London theory to provide an understanding of magnetism of these carbon clusters [35]. These studies along with earlier experimental measurements of the magnetic susceptibility on fullerenes [36,37] confirm the cancelation of the diamagnetism by a very strong Van Vleck paramagnetic term, giving rise to a vanishingly small π -electron ring current magnetic susceptibility in these systems.

Conclusions

There are at least three singlet excitations below the lowest optically allowed excitation (oscillator strength 0.05) in pyracylene within the PPP model. It has been shown that in the ground state pyracylene has very strong ethylenic units on the pentagons. These are weakly bridge to a distorted naphthalenic unit with rather long pentagon–hexagon bonds.

Studies on fluoranthene show that this molecule could be viewed as weakly coupled benzene and naphthalene molecules. Studies on corannulene show the lowest excitation to be at 3.93 eV for uniform bond lengths. Introduction of longer

hexagon–pentagon bonds leads to a blue shift in the absorption to 4.18 eV. The correlated hexagon–hexagon bonds have a bond order (0.62) which is nearly the same as in benzene (0.69). Hexagon–pentagon bond order is lower (0.46) while the outer hexagon bonds have strong ethylenic character (bond order ≈ 0.79). The hexagon–pentagon and hexagon–hexagon bonds in the similar environments in pyracylene and fluoranthene are also similar in corannulene.

The ground state ring currents in pyracylene and fluoranthene molecules show interesting variations. Pyracylene has paramagnetic circulation along the perimeter while the naphthalenic unit has diamagnetic circulation. In fluoranthene the five-membered ring shows a very weak diamagnetic ring current associated with a slight negative charge on this ring.

The bond lengths (obtained from bond order data using Coulson's formula [31,32]) of the hexagon–hexagon and hexagon–pentagon bonds in the interior of these molecules are consistent with experimental bond lengths for the C_{60} molecule. It is interesting to note that the optical absorption of C_{60} which spans a wide energy range can be viewed as though the low-energy excitations are localized on pyracylene type fragments while the high energy excitations are confined to fluoranthene and corannulene type fragments.

References

- [1] H.W. Kroto, J.R. Heath, S.C. O'Brien, R.F. Curl and R.E. Smalley, *Nature*, 318 (1985) 162.
- [2] W. Kratschmer, L.D. Lamb, L.K. Fostiropoulos and D.R. Huffman, *Nature*, 347 (1990) 354.
- [3] G. Baskaran and E. Tosatti, *Curr. Sci.*, 61 (1991) 33.
- [4] S. Chakravarthy, M.P. Gelfand and S. Kivelson, *Science*, 254 (1991) 970.
- [5] W.I.F. David, R.M. Ibberson, J.C. Matthewman, K. Prassides, T.J.S. Denis, J.P. Hare, H.W. Kroto, R. Taylor and D.R.M. Walton, *Nature*, 353 (1991) 147.
- [6] R.M. Fleming, A.P. Ramirez, M.J. Rosseinsky, D.W. Murphy, R.C. Haddon, S.M. Zahurak and A.V. Makhija, *Nature*, 352 (1991) 787.
- [7] P. Allemand, K.C. Khemani, A. Koch, F. Wudl, K. Holczer, S. Donovan, G. Gruner and J.E. Thompson, *Science*, 253 (1991) 301.
- [8] P.W. Stephens, D. Cox, J.W. Lauher, L. Mihaly, J.B. Wiley, P.M. Allemand, A. Hirsch, K. Holczer, Q. Li, J.D. Thompson and F. Wudl, *Nature*, 355 (1992) 331.

- [9] C.M. Varma, J. Zaanen and K. Raghavachari, *Science*, 254 (1991) 989.
- [10] M. Schuster, M. Lannoo, M. Needels, G.A. Baraff and D. Tomanek, *Phys. Rev. Lett.*, 68 (1992) 526.
- [11] C.C. Zhang and C.M. Lieber, *Science*, 254 (1991) 1619.
- [12] S. Chakravarty, S. Khlebnikov and S. Kivelson, *Phys. Rev. Lett.*, 69 (1992) 212.
- [13] M.L. Cohen and P.W. Anderson, *AIP Conf. Proc.*, 4 (1972) 17.
- [14] J.M. Schulman, R.L. Disch, M.A. Miller and R.C. Peck, *Chem. Phys. Lett.*, 141 (1987) 45.
- [15] D. Bakowiesand and W. Thiel, *J. Am. Chem. Soc.*, 113 (1991) 3704.
- [16] S. Sathpathy, *Chem. Phys. Lett.*, 130 (1986) 545.
- [17] M.J. Rosseninsky, A.P. Ramirez, S.H. Glarum, D.W. Murphy, R.C. Haddon, A.F. Hebard, T.T.M. Palstra, A.R. Kortan, S.M. Zahurak and A.V. Makhija, *Phys. Rev. Lett.*, 66 (1991) 2830.
- [18] P.J. Fagan, J.C. Calabrese and B. Malone, *Science*, 252 (1991) 1160.
- [19] D. Amic and N. Trinajstić, *J. Chem. Soc., Perkin, Trans. 2*, (1995) 1990.
- [20] J. Hubbard, *Proc. R. Soc. London, Ser. A*, 276 (1964) 238; 277 (1964) 237; 281 (1964) 401.
- [21] R. Pariser and R.G. Parr, *J. Chem. Phys.* 21 (1953) 466; J.A. Pople, *Trans. Faraday Soc.* 49 (1953) 1375.
- [22] Z.G. Soos and S. Ramasesha, in D.J. Klein and N. Trinajstić (Eds.), *Valence Bond Theory and Chemical Structure*, Elsevier, Amsterdam, 1990, p. 81.
- [23] B. Srinivasan and S. Ramasesha, *Solid State Commun.* 81 (1992) 831. S. Ramasesha and B. Srinivasan, *Indian J. Chem.*, 31A,B (1992) F56.
- [24] Y. Anusooya and S. Ramasesha, unpublished results.
- [25] K. Ohno, *Theor. Chim. Acta.* 2 (1964) 219.
- [26] S. Rettrup, *J. Comput. Phys.*, 45 (1982) 100.
- [27] S. Kuwajima and Z.G. Soos, *J. Am. Chem. Soc.*, 109 (1987) 107.
- [28] B.M. Trost, G.M. Bright, C. Frihart and D. Brittelli, *J. Am. Chem. Soc.*, 93 (1971) 737.
- [29] J. Kolc, E.W. Thulstrup and J. Michl, *J. Am. Chem. Soc.*, 96 (1974) 7188.
- [30] K. Hedberg, L. Hedberg, D.S. Bethune, C.A. Brown, H.C. Dorn, R.D. Johnson and M. de Vries, *Science*, 254 (1991) 410.
- [31] A. Streitwieser, Jr., *Molecular Orbital Theory for Organic Chemists*, Wiley, New York, 1961, p. 167.
- [32] C.A. Coulson, *Proc. R. Soc. London, Ser. A*, 169 (1939) 413.
- [33] C.A. Coulson and R.B. Mallion, *J. Am. Chem. Soc.*, 98 (1976) 592.
- [34] J.A.N.F. Gomes and R.B. Mallion, *J. Org. Chem.*, 46 (1981) 719.
- [35] R.C. Haddon and A. Pasquarello, *Phys. Rev. B*, 50 (1994) 16459.
- [36] R.C. Haddon, L.F. Schneemeyer, J.V. Waszczak, S.H. Glarum, R. Tycko, G. Dabbagh, A.R. Kortan, A.J. Muller, A.M. Muijsce, M.J. Rosseinsky, S.M. Zahurak, A.V. Makhija, F.A. Thiel, K. Raghavachari, E. Cockayne and V. Elser, *Nature*, 350 (1991) 46.
- [37] R.S. Ruoff, D. Beach, J. Cuomo, T. McGuire, R.L. Whetten and F. Diedrich, *J. Phys. Chem.*, 95 (1991) 3457.
- [38] J.P. Hare, H.W. Kroto and R. Taylor, *Chem. Phys. Lett.*, 177 (1991) 394.